Classtest-3. Digital Logic Design. Time- 30 min. Marks- 20.

Answer the following questions:

1.	Differentiate synchronous and asynchronous counters with necessary diagrams. Draw a circuit diagram of a MOD-6 synchronous counter with truth table and input expression.	10
2.	Draw the circuit diagram of a MOD-9 up/down synchronous counter with truth table and input expression. Draw the timing diagram for the given up/down input=1111001111000000011111000000.	10